

Hollow Prototype Infrastructure Audit

M0-M30 current-state review, validation evidence, risk register, and next milestone recommendations

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Executive Summary

- M24 platform QA is currently Failed: Addressables passed, but dependency audit failed, Unity test suites were recorded as separate NotRun commands, Windows build support is missing, and VisionOS simulator tooling is incomplete.
- M25 vertical slice lock is currently Failed: branch content, hub/shop/portals, ArtPass lock, and platform checklist passed, but upstream M0-M24 validation failed.
- Current validation debt is concentrated around historical branch topology validators: M14, M15, M17, plus M18 reward-pool scene wiring expectations after later reward systems superseded M18.
- M27, M28, M29, and M30 validators pass individually after the latest work; M26 has tests but lacks a formal milestone doc/generator/validator wrapper.
- Git state after refreshing gates contains 13 generated report/PDF changes; these should be committed intentionally or cleaned after review.

Category	Current Status	Notes/Recommendations
M24 dependency audit	20/24 passed	Core editor validation is not green until M14/M15/M17/M18 are fixed or updated.
M24 QA result	Failed	Build/device handoff is not ready; some blocks are environment-only.
M25 lock result	Failed	Slice content is mostly coherent but cannot be called locked while upstream validators fail.
Script assemblies	16	Architecture is modular enough for continued milestone work.
Scenes	6	Expected platform/menu/designer scene split exists.
EditMode milestone tests	31	Broad milestone test coverage exists, including M30.
PlayMode smoke files	2	PlayMode smoke coverage is present but sparse.
Dirty files after audit	13	Audit refreshed generated evidence; review before committing.

Current Infrastructure Map

The project has the intended Hollow-owned Unity structure: modular script assemblies, separate gameplay/platform/designer scenes, generated content assets, report outputs, and milestone test files. The shape is strong; the immediate issue is validation consistency and release-readiness, not lack of systems.

Scenes	Boot.unity, Game_VisionOS_Bounded.unity, Game_VisionOS_Immersive.unity, Game_Windows.unity, MainMenu.unity, RoomDesigner.unity	Boot, menu, Windows, Vision Pro bounded, Vision Pro immersive, and Room Designer are present.
Assemblies	16 asmdefs under Assets/_Hollow/Scripts	Matches the intended split across core, platform, input, UI, rooms, branches, combat, rewards, persistence, presentation, diagnostics, editor, and designer.
Content data	Rooms, macro fixtures, approved designer intake, weapons, rewards, characters, equipment, synergies, encounters, presentation, platform QA.	The project has evolved beyond the M25 lock; M30 content exists and should be considered current baseline after cleanup.
Testing	31 EditMode milestone test files and 2 PlayMode smoke files.	Coverage breadth is good, but M24 records test commands rather than executing them inside the QA gate.
Generated reports	latest_m24_dependency_audit, latest_platform_build_qa, latest_vertical_slice_lock refreshed.	Evidence is current, but report files are dirty and should be intentionally committed or excluded.

Primary architectural conclusion: the prototype has a rich runtime spine - branch generation, imported rooms, Room Designer, combat, rewards, shops, ArtPass, platform shells, characters, weapons, items/cards, armor, and synergies. The next phase should stabilize validator contracts and QA automation before adding another wide layer of systems.

Milestone Status Matrix

Status combines fresh validator evidence, static repo inspection, and current reports. "Passed" means the available milestone validator passed. "Failed" means current validator/gate evidence failed. "Coverage Gap" means the feature exists but lacks the expected milestone gate wrapper.

0	Milestone 0 - Unity Foundation	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
1	Milestone 1 - Menu, Profiles, And Platform Routing	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
2	Milestone 2 - Shared Runtime Shell	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
3	Milestone 3 - HollowRuntime V2 Import	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
4	Milestone 4 - Playable Combat Loop	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
5	Milestone 5 - Enemy Archetypes and Diagnostics	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
6	Milestone 6 - Branch Traversal, Rewards, and Hub Return	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
7	Milestone 7: Run Economy, Persistence, Meta Progression	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
8	Milestone 8: Room Designer Mode	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
9	Milestone 9: Content Pipeline, Materials, VFX, Audio, Addressables	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
10	Milestone 10: Platform Polish	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
11	Milestone 11: Prototype Lock	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
12	Milestone 12: Build Automation And Full Prototype Audit	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
13	Milestone 13: Macro-Room Runtime Foundations	Passed	M24 dependency audit: OK	Stable foundation. Keep green while repairing later branch/reward validators.
14	Milestone 14: Seeded Macro Branch Generation	Failed	M24 dependency audit: M14 macro graph must have five rooms, eight directed connections, and twelve occupied branch cells.; M14 west block room must not connect through east_0; only east_1 is connected.	Update legacy exact-count assertions for Isaac-like auto-connect rules or preserve a separate M14 fixture validator path.
15	Milestone 15: Seeded Procedural Macro Branches	Failed	M24 dependency audit: M15 seeded graph must have eight rooms, fourteen directed explicit-port connections, and the M15 branch identity.	Rebaseline seeded macro branch validator around invariants: 8 rooms, connectivity, boss endpoint, no overlaps, explicit ports, expected branch ID per current generator.
16	Milestone 16: Approved Room Designer Pool	Passed	M24 dependency audit: OK	Keep as implemented; retain validator in regression chain.
17	Milestone 17: Feature Branch Treasure Rooms	Failed	M24 dependency audit: M17 feature graph must keep eight rooms, explicit directed port connections, and the M17 branch identity.; M17 treasure room should be selected as a leaf when the generated topology allows it.	Update feature-room validator for current topology rules or pin a historical M17 generator fixture for regression compatibility.
18	Milestone 18: Seeded Random Rewards	Failed	M24 dependency audit: Assets/_Hollow/Scenes/Game_Windows.unity BranchSessionController is not wired to M18 reward pools.; Assets/_Hollow/Scenes/Game_VisionOS_Bounded.unity BranchSessionController is not wired to M18 reward pools.; Assets/_Hollow/Scenes/Game_VisionOS_Immersive.unity BranchSessionController is not wired to M18 reward pools.	Make reward-pool scene wiring validation version-aware; M30 pools supersede M18 pools but must still satisfy seeded reward invariants.
19	Milestone 19: Enemy + Encounter Content V1	Passed	M24 dependency audit: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
20	Milestone 20: Branch Features, Shops, Secrets, Keys/Locks	Passed	M24 dependency audit: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
21	Milestone 21: Shop Choice UI + Purchasable Rewards	Passed	M24 dependency audit: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.

22	Milestone 22: Room Designer Macro Authoring Polish	Passed	M24 dependency audit: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
23	Milestone 23: Art/Content Replacement Pipeline	Passed	M24 dependency audit: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
24	Milestone 24: Platform Build And Device QA	Mixed	M24 QA + M25 lock: Milestone24Validator passed inside M25 lock run, but full M24 QA gate failed: dependency audit failed, tests not run by gate, Windows and VisionOS were environment-blocked.	Fix upstream validator failures, run separate EditMode/PlayMode test commands, install Windows build support and VisionOS simulator tooling.
25	Milestone 25: Vertical Slice Content Lock	Failed	Fresh M25 lock gate: Vertical slice branch/content, hub/shop, ArtPass lock, and platform checklist passed; m0-m24 audit failed upstream.	Rerun after M14/M15/M17/M18 are green; lock should then pass because slice content checks already pass.
26	Milestone 26	Coverage Gap	Static repo audit: EditMode test exists, but no Milestone26 docs, generator, or validator were found. Treat M26 as implemented foundation work without milestone gate coverage.	Add M26 doc, generator if needed, and validator or formally merge M26 into M27-M30 build-foundation track.
27	Milestone 27: Weapon Mode + Light/Heavy Attacks	Passed	Fresh M27 validator: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
28	Milestone 28: Items, Cards, Coins, And Shop Rewards	Passed	Fresh M28 validator: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
29	Milestone 29: Characters + Passive Identity Skills	Passed	Fresh M29 validator: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.
30	Milestone 30: Synergy Tags + Starter Build Variety	Passed	Fresh M30 validator: OK	Current implementation validates; keep in current baseline and ensure older validators stop conflicting with it.

Validation, Build, and Device Evidence

m0-m23-audit	Failed	M0-M23 audit failed: 4/24.	Open latest_m24_dependency_audit.md for failing validator details.
addressables-build	Passed	Local Addressables content built successfully.	OK
editmode-tests	NotRun	Run EditMode tests with `Unity -batchmode -projectPath <repo> -runTests -testPlatform editmode -testResults output/reports/m24-editmode-results.xml`.	Run this command as a separate Unity Test Runner invocation; the in-process QA gate records the expected command and the PlayMode/EditMode suites remain independently verifiable.
playmode-smoke-tests	NotRun	Run PlayMode smoke tests with `Unity -batchmode -projectPath <repo> -runTests -testPlatform playmode -testResults output/reports/m24-playmode-results.xml`.	Run this command as a separate Unity Test Runner invocation; the in-process QA gate records the expected command and the PlayMode/EditMode suites remain independently verifiable.
windows-development-build	BlockedByEnvironment	Standalone Windows 64-bit build support is not installed in this Unity editor.	Install Unity Windows Build Support for Unity 6000.4.1f1.; Rerun Hollow/Platform QA/Build Windows Development M24.
visionos-readiness	BlockedByEnvironment	Vision Pro project readiness is present, but local simulator/build tooling is incomplete.	Install Xcode command line tools and verify `xcrun simctl` works for simulator QA.

Vertical Slice Lock Evidence

lock-definition	Passed	M25 lock asset pins branch identity, seed, catalogs, platform QA, and slice counts.	OK
branch-content	Passed	Locked branch generated with 8 rooms, 18 directional connections, 5 fixtures, and 0 approved rooms.; Boss key source: room_02; secret: room_06; boss: boss_01.	OK
hub-shop-portals	Passed	Inter-branch hub exposes three shop offers and three seeded next-branch portal choices.	OK
artpass-lock	Passed	Required ArtPass roles/cues resolved without prototype fallback. Warnings: 0.	OK
platform-checklist	Passed	Windows, Vision Pro bounded, and Vision Pro immersive have equal vertical-slice checklist coverage.	OK
m0-m24-audit	Failed	M0-M24 validators failed: 4/25.	Milestone14Validator; Milestone15Validator; Milestone17Validator; Milestone18Validator

Key Risks and Suggested Fixes

P0	Historical branch validators fail after later topology changes	M14/M15/M17 failures block M24/M25 despite current branch systems passing later validators.	Create a validator policy: historical validators assert durable invariants, while current branch validators own exact topology. Patch M14/M15/M17 accordingly and add explicit tests for auto-connect, boss endpoint, and secret single-room rules.
P0	M18 reward-pool wiring validator conflicts with newer reward pools	M30 reward pools replace and extend M18 behavior, so strict M18 scene references report false failure.	Update M18 validator to accept M18+ compatible reward pools or verify behavior by reward kinds/effects rather than exact asset references.
P0	M24 QA gate does not run tests in-process	The gate reports EditMode/PlayMode as NotRun, so build handoff cannot claim complete automated test coverage.	Add shell/batch orchestration that runs Unity Test Runner separately, captures XML, parses pass/fail, and folds results into M24 report.
P1	Platform build/device environment incomplete	Windows build support and VisionOS simulator tooling are blocked locally.	Install Unity Windows Build Support, verify Xcode/simctl, rerun M24. Keep environment blocks acceptable only with remediation text.
P1	M26 process gap	Player build foundation exists in tests/runtime, but no milestone doc/generator/validator makes audit lineage fuzzy.	Backfill M26 docs and a lightweight validator that checks save models, wallet/equipment/inventory state, and HUD summaries.
P2	Generated report churn	Running gates mutates latest reports and M25 PDF, creating noisy diffs.	Adopt a report artifact policy: commit milestone evidence intentionally, or route timestamped local QA outputs to ignored paths except stable handoff PDFs.

Suggested Next 10 Milestones: M31-M40

M31	Validation Debt Recovery + Clean Prototype Baseline	Make M14/M15/M17/M18 green under current branch/reward rules; backfill M26 wrapper; commit or clean generated evidence.	M0-M30 validators green or explicitly retired; clean git baseline.
M32	Full QA Gate Rebaseline + Test Runner Reliability	Make M24 run/parse EditMode and PlayMode XML, report failures clearly, and document local environment prerequisites.	M24 gate reports test XML results and environment blocks only where genuine.
M33	Combat Feel, Physics, Collision, and Camera Polish	Tune movement, collision, hit feedback, enemy grace, projectile readability, camera follow, and VisionOS comfort without new systems.	Manual smoke confirms movement/combat/camera feel better on Windows and VisionOS routes.
M34	Shield / Defense / Armor Behavior V1	Turn defense/armor into visible gameplay: block, parry/push prototype, stamina cost, enemy interaction, and HUD feedback.	Shield/armor interactions are testable, saved safely, and readable in HUD.
M35	Challenge Mode V1	Add transient curated challenges from main menu: fixed seeds, weapon-limited, shop-limited, boss, secret, and room-designer challenge routes.	Challenge entries launch transient sessions and never mutate profile runs unless intended.
M36	Room and Encounter Content Expansion	Approve a small branch-ready room pool, add more encounter recipes, improve secret/boss/key room variety, keep seeded determinism.	Approved room pool grows and branch generation remains deterministic and valid.
M37	Enemy/Boss Behavior Readability Pass	Improve telegraphs, spawn pacing, attack cooldowns, boss phases, projectile contrast, and difficulty bands.	Enemy/boss encounters pass readability tests and do not spike unfair damage.
M38	ArtPass Integration Sprint with Rafal Asset Pipeline	Replace core visuals through presentation roles, validate no visual colliders/scripts, add VFX/audio placeholders, document asset intake checklist.	Required ArtPass bindings resolve without prototype fallback for core slice roles.
M39	Story, World Identity, and Run Framing V1	Add run-start framing, world/floor naming, boss/secret/shop lore prompts, and minimal narrative UI that does not block gameplay.	Players understand why they enter branches, hubs, shops, keys, and bosses.
M40	Vertical Slice Re-Lock + External Handoff Build	Rerun full M24/M25 gates, produce Windows dev build when environment supports it, update manual QA checklist, and publish handoff PDF/package.	M24 and M25 are green or only environment-blocked; handoff artifact is ready.

Appendix: Evidence and Generated Artifacts

Fresh evidence files generated or refreshed during this audit run:

- output/reports/latest_m24_dependency_audit.json / .md - M0-M23 dependency audit, currently 20/24 passed.
- output/reports/latest_platform_build_qa.json / .md - M24 platform QA, currently Failed with environment blocks.
- output/reports/latest_vertical_slice_lock.json / .md - M25 vertical slice lock, currently Failed due upstream audit.
- output/pdf/Hollow_M25_Verical_Slice_Content_Lock.pdf - regenerated by the M25 gate.
- This PDF - output/pdf/Hollow_Prototype_Infrastructure_Audit_M0_M30_and_Next_Milestones.pdf.

Known Environment Limitations

- Unity licensing log reports an unavailable access token during batchmode, but validators still executed.
- Windows Standalone build support is not installed in this Unity editor, so Windows player build is BlockedByEnvironment.
- VisionOS readiness is present, but local simulator/build tooling is incomplete, so device/simulator QA is BlockedByEnvironment.
- M24 records EditMode and PlayMode test commands as NotRun; separate test runner orchestration is recommended in M32.

Current Git State After Audit

13 changed/untracked files are currently present after gate refresh. They are primarily latest reports, timestamped reports, and regenerated M25 PDF output.

```
M output/pdf/Hollow_M25_Verical_Slice_Content_Lock.pdf
M output/reports/latest_m24_dependency_audit.json
M output/reports/latest_m24_dependency_audit.md
M output/reports/latest_platform_build_qa.json
M output/reports/latest_platform_build_qa.md
M output/reports/latest_vertical_slice_lock.json
M output/reports/latest_vertical_slice_lock.md
?? output/reports/platform-build-qa-20260427-232900.json
?? output/reports/platform-build-qa-20260427-232900.md
?? output/reports/prototype-audit-20260427-232900.json
?? output/reports/prototype-audit-20260427-232900.md
?? output/reports/vertical-slice-lock-20260427-233008.json
?? output/reports/vertical-slice-lock-20260427-233008.md
```